

LightBandit Fabrication Document

Layer Stack Legend

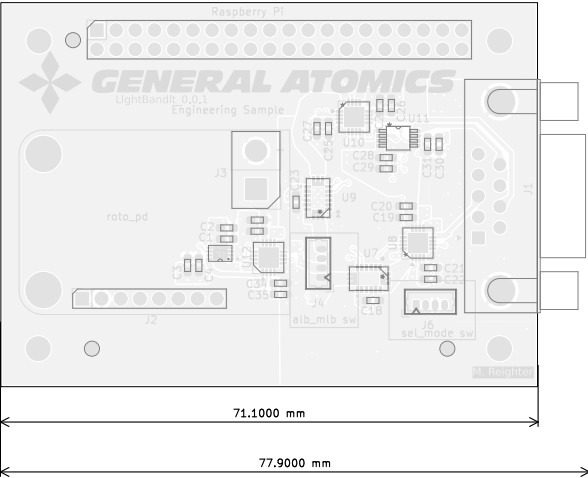
|  | Material     | Layer         | Thickness        | Dielectric    | Type        | Gerber |
|--|--------------|---------------|------------------|---------------|-------------|--------|
|  | F,Paste      |               |                  |               | Paste Mask  |        |
|  | F,Silkscreen |               |                  |               | Legend      | GBR    |
|  | F,Mask       |               | 0,02mm           | Solder Resist | Solder Mask | GBR    |
|  | Copper       | L1 (Sig, PWR) | 0,07mm (2,00oz)  |               | Signal      | GBR    |
|  | Prepreg      |               | 0,1mm            | FR4_7628      | Dielectric  |        |
|  | Copper       | L2 (GND)      | 0,035mm (1,00oz) |               | Plane       | GBR    |
|  | Core         |               | 1,21mm           | FR4           | Dielectric  |        |
|  | Copper       | L3 (Sig, PWR) | 0,035mm (1,00oz) |               | Signal      | GBR    |
|  | Prepreg      |               | 0,1mm            | FR4_7628      | Dielectric  |        |
|  | Copper       | L4 (Sig, PWR) | 0,07mm (2,00oz)  |               | Signal      | GBR    |
|  | B,Mask       |               | 0,02mm           | Solder Resist | Solder Mask | GBR    |
|  | B,Silkscreen |               |                  |               | Legend      | GBR    |
|  | B,Paste      |               |                  |               | Paste Mask  |        |

Total thickness: 1,66mm  
Note: external layer thicknesses are specified after plating

Impedance Table

| Transmission Line              | Impedance [ohms] | Tolerance [ohms] | Layer | Trace Width [mm] | Gap [mm] | Ref. Layers |
|--------------------------------|------------------|------------------|-------|------------------|----------|-------------|
| Edge-Coupled Coated Microstrip | 100              | ±10 %            | L1    | 0,2032           | 0,28     | L2          |

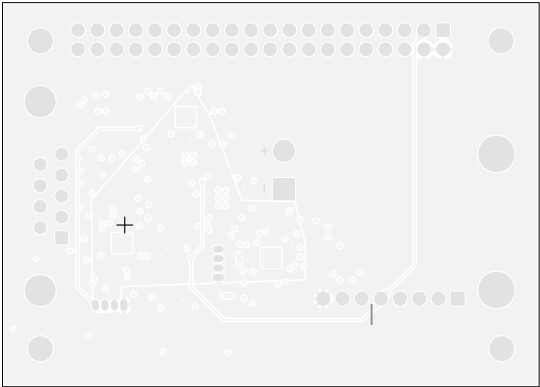
Top Fabrication (Scale 1:1)



# LightBandit Fabrication Document

3 × 120°

## Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

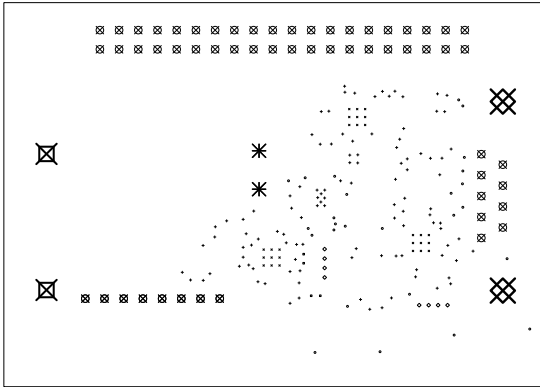
|  |                                                |                                   |                              |                              |                             |
|--|------------------------------------------------|-----------------------------------|------------------------------|------------------------------|-----------------------------|
|  | Comments:                                      | Company:<br>General Atomics EMS   |                              | Variant:<br>PRELIMINARY      | Git Hash:<br>1b1e0f8        |
|  |                                                | Board Name:<br>LightBandit        |                              | Project Name:<br>LightBandit |                             |
|  | Sheet Title:<br>Bottom Fabrication (Scale 1:1) | File Name:<br>proalbm1b.kicad_pcb | Designer:<br>Marcus Reighter | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                    |                                   | Reviewer:                    | Size:<br>A4                  | Sheet:<br>2 of 10           |

LightBandit Fabrication Document

Drill Table

| Symbol    | Count | Hole Size           | Plated | Hole Shape | Drill Layer Pair              | Hole Type |
|-----------|-------|---------------------|--------|------------|-------------------------------|-----------|
| ×         | 27    | 0,20mm (7,87mils)   | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad       |
| ○         | 26    | 0,25mm (9,84mils)   | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Via       |
| +         | 110   | 0,30mm (11,81mils)  | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Via       |
| □         | 4     | 0,30mm (11,81mils)  | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad       |
| ◇         | 8     | 0,50mm (19,69mils)  | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad       |
| ⊠         | 65    | 1,00mm (39,37mils)  | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad       |
| ✱         | 4     | 1,75mm (68,90mils)  | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad       |
| ⊠         | 2     | 2,70mm (106,30mils) | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad       |
| ⊠         | 2     | 3,20mm (125,98mils) | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad       |
| Total 248 |       |                     |        |            |                               |           |

Drill Drawing L1 - L4 (Scale 1:1)



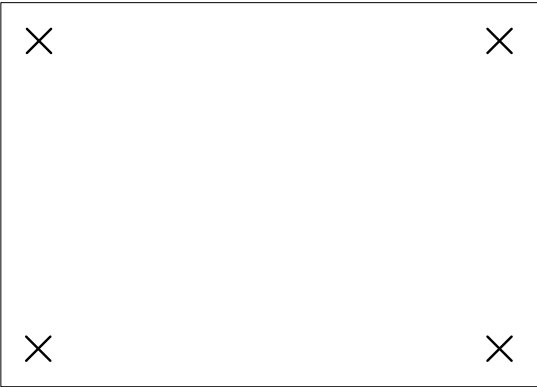
|  |                                         |                                   |                              |                              |                             |
|--|-----------------------------------------|-----------------------------------|------------------------------|------------------------------|-----------------------------|
|  | Comments:                               | Company:<br>General Atomics EMS   |                              | Variant:<br>PRELIMINARY      | Git Hash:<br>1b1e0f8        |
|  |                                         | Board Name:<br>LightBandit        |                              | Project Name:<br>LightBandit |                             |
|  | Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>proalbm1b.kicad_pcb | Designer:<br>Marcus Reighter | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                             |                                   | Reviewer:                    | Size:<br>A4                  | Sheet:<br>3 of 10           |

# LightBandit Fabrication Document

Drill Table

| Symbol | Count   | Hole Size           | Plated | Hole Shape | Drill Layer Pair              | Hole Type  |
|--------|---------|---------------------|--------|------------|-------------------------------|------------|
| X      | 4       | 3.20mm (125,98mils) | NPPTH  | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Mechanical |
|        | Total 4 |                     |        |            |                               |            |

Drill Drawing L1 - L4 (Scale 1:1)



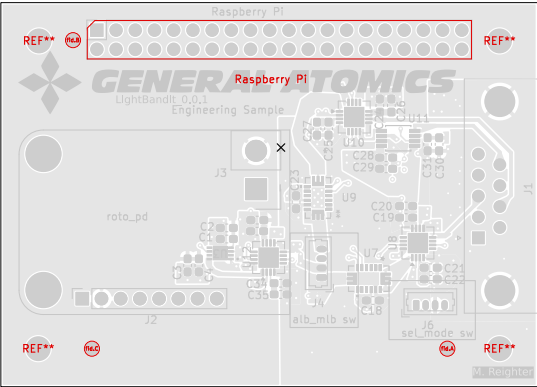
|  |                                         |                                   |                              |                              |                             |
|--|-----------------------------------------|-----------------------------------|------------------------------|------------------------------|-----------------------------|
|  | Comments:                               | Company:<br>General Atomics EMS   |                              | Variant:<br>PRELIMINARY      | Git Hash:<br>1b1e0f8        |
|  |                                         | Board Name:<br>LightBandit        |                              | Project Name:<br>LightBandit |                             |
|  | Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>proalbm1b.kicad_pcb | Designer:<br>Marcus Reighter | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                             |                                   | Reviewer:                    | Size:<br>A4                  | Sheet:<br>4 of 10           |

# LightBandit Fabrication Document

Top Test Points (Scale 1:1)

| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|

| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|



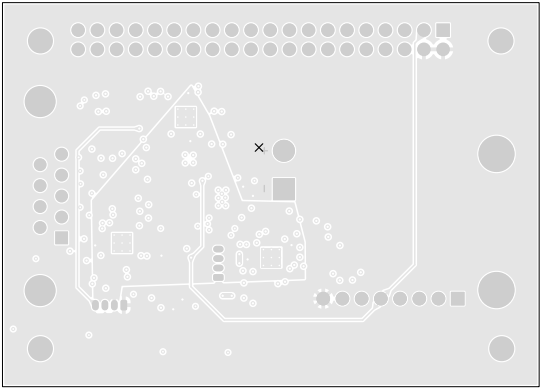
All dimensions are in millimeters unless otherwise specified.

|  |                                             |                                   |                              |                              |                             |
|--|---------------------------------------------|-----------------------------------|------------------------------|------------------------------|-----------------------------|
|  | Comments:                                   | Company:<br>General Atomics EMS   |                              | Variant:<br>PRELIMINARY      | Git Hash:<br>1b1e0f8        |
|  |                                             | Board Name:<br>LightBandit        |                              | Project Name:<br>LightBandit |                             |
|  | Sheet Title:<br>Top Test Points (Scale 1:1) | File Name:<br>proalbm1b.kicad_pcb | Designer:<br>Marcus Reighter | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                 |                                   | Reviewer:                    | Size:<br>A4                  | Sheet:<br>5 of 10           |

# LightBandit Fabrication Document

Bottom Test Points (Scale 1:1)

| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|

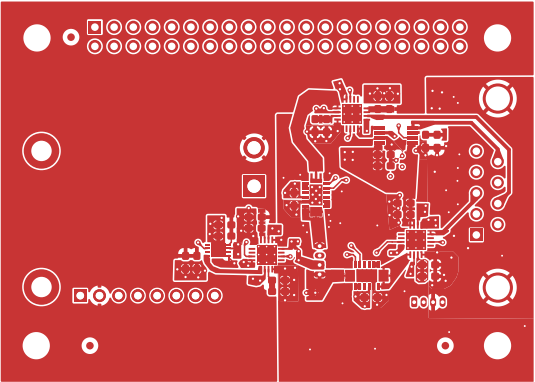


All dimensions are in millimeters unless otherwise specified.

|  |                                                |                                   |                              |                              |                             |
|--|------------------------------------------------|-----------------------------------|------------------------------|------------------------------|-----------------------------|
|  | Comments:                                      | Company:<br>General Atomics EMS   |                              | Variant:<br>PRELIMINARY      | Git Hash:<br>1b1e0f8        |
|  |                                                | Board Name:<br>LightBandit        |                              | Project Name:<br>LightBandit |                             |
|  | Sheet Title:<br>Bottom Test Points (Scale 1:1) | File Name:<br>proalbm1b.kicad_pcb | Designer:<br>Marcus Reighter | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                    |                                   | Reviewer:                    | Size:<br>A4                  | Sheet:<br>6 of 10           |

# LightBandit Fabrication Document

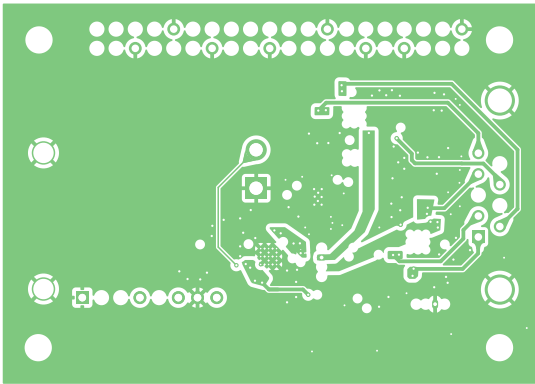
L1 (Sig, PWR) (Scale 1:1)



|  |                           |                     |                 |               |                |
|--|---------------------------|---------------------|-----------------|---------------|----------------|
|  | Comments:                 | Company:            |                 | Variant:      | Git Hash:      |
|  |                           | General Atomics EMS |                 | PRELIMINARY   | 1b1e0f8        |
|  | Sheet Title:              | Board Name:         |                 | Project Name: |                |
|  |                           | LightBandit         |                 | LightBandit   |                |
|  | L1 (Sig, PWR) (Scale 1:1) | File Name:          | Designer:       | Date:         | Revision:      |
|  |                           | proalbmlb.kicad_pcb | Marcus Reighter | 2024-04-13    | + (Unreleased) |
|  | Sheet Path:               |                     | Reviewer:       | Size:         | Sheet:         |
|  |                           |                     |                 | A4            | 7 of 10        |

# LightBandit Fabrication Document

L2 (GND) (Scale 1:1)

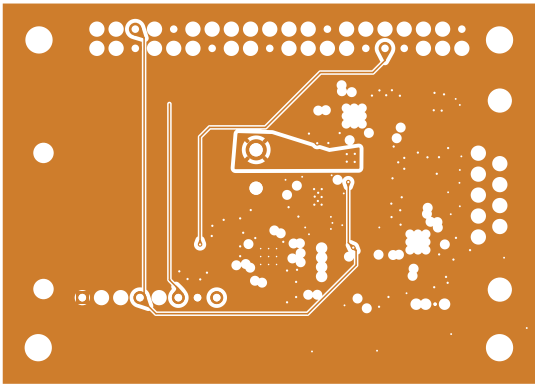


|  |                                      |                                   |                              |                              |                             |
|--|--------------------------------------|-----------------------------------|------------------------------|------------------------------|-----------------------------|
|  | Comments:                            | Company:<br>General Atomics EMS   |                              | Variant:<br>PRELIMINARY      | Git Hash:<br>1b1e0f8        |
|  |                                      | Board Name:<br>LightBandit        |                              | Project Name:<br>LightBandit |                             |
|  | Sheet Title:<br>L2 (GND) (Scale 1:1) | File Name:<br>proalbm1b.kicad_pcb | Designer:<br>Marcus Reighter | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                          |                                   | Reviewer:                    | Size:<br>A4                  | Sheet:<br>8 of 10           |



# LightBandit Fabrication Document

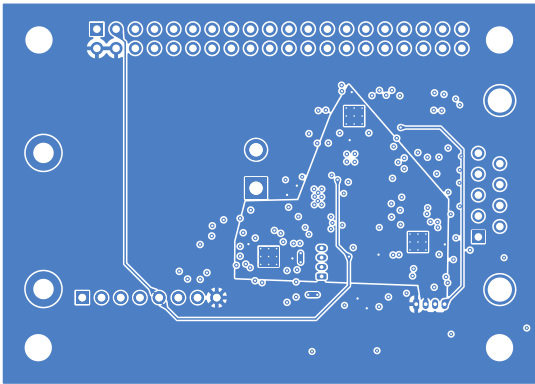
## L3 (Sig, PWR) (Scale 1:1)



|  |                                           |                                   |                              |                              |                             |
|--|-------------------------------------------|-----------------------------------|------------------------------|------------------------------|-----------------------------|
|  | Comments:                                 | Company:<br>General Atomics EMS   |                              | Variant:<br>PRELIMINARY      | Git Hash:<br>1b1e0f8        |
|  |                                           | Board Name:<br>LightBandit        |                              | Project Name:<br>LightBandit |                             |
|  | Sheet Title:<br>L3 (Sig, PWR) (Scale 1:1) | File Name:<br>proalbmlb.kicad_pcb | Designer:<br>Marcus Reighter | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                               |                                   | Reviewer:                    | Size:<br>A4                  | Sheet:<br>9 of 10           |

# LightBandit Fabrication Document

L4 (Sig, PWR) (Scale 1:1)



|  |                                           |                                   |                              |                              |                             |
|--|-------------------------------------------|-----------------------------------|------------------------------|------------------------------|-----------------------------|
|  | Comments:                                 | Company:<br>General Atomics EMS   |                              | Variant:<br>PRELIMINARY      | Git Hash:<br>1b1e0f8        |
|  |                                           | Board Name:<br>LightBandit        |                              | Project Name:<br>LightBandit |                             |
|  | Sheet Title:<br>L4 (Sig, PWR) (Scale 1:1) | File Name:<br>proalbmlb.kicad_pcb | Designer:<br>Marcus Reighter | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                               |                                   | Reviewer:                    | Size:<br>A4                  | Sheet:<br>10 of 10          |